Sales Performance Analysis

End-to-End Analysis Case Study | Data Analyst Course, Week 4 Capstone Project

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Summary for Stakeholders

This report analyses 700 sales orders from September 2013 to December 2014, covering 5 customer segments, 5 countries and 6 products. The business earned **118.73M** in sales and **16.89M** in profit, a profit margin of **14.23%**. Sales are growing, but two problems are holding profit back: **heavy discounting** and a **loss-making Enterprise segment**.

118.73M	16.89M	14.23%	58
Total Sales	Total Profit	Profit Margin	Loss-making Orders

Key findings

- **Discounts reduce profit.** The profit margin is 21.9% on orders with no discount, but only 9.1% on orders with a high discount.
- **Enterprise loses money.** All 58 loss-making orders are Enterprise orders sold with a High (33) or Medium (25) discount. Together they lost 777,321, which is 4.6% of total profit.
- **Government drives profit.** It earns 11.39M, which is 67% of all profit. Channel Partners has the best margin (73%) but the smallest sales (1.80M).
- **Sales are growing.** September to December 2014 sales are 37% higher than the same months of 2013, and 2014 Q4 is the best quarter (29.76M).

What we recommend

- Require approval for Medium and High discounts on Enterprise orders, because none of those orders made a profit.
- Review the overall discount policy, because bigger discounts lower the margin in every band.
- Test how to grow the high-margin Channel Partners and Midmarket segments, and protect the Government segment.

1. Project Overview

The goal of this project is to take a raw sales dataset through the full analysis process: clean it, analyse it, visualise the results and recommend actions. The work was done with the following tools.

Stage	Tool	What was done
Clean and explore	Python (Pandas)	Loaded the data, fixed formats, checked quality and added new columns
Analyse	Python (Pandas)	Summarised sales, profit and margin by segment, discount band, product, country and quarter
Visualise	Excel	Pivot tables, pivot charts, KPI cells and slicers on a dashboard
Visualise	Power BI	Measures, KPI cards, charts and slicers on an interactive dashboard
Report	This document	Findings, charts, recommendations and a stakeholder summary

About the dataset

- 700 orders and 16 original columns, from 01 September 2013 to 01 December 2014.
- Columns include Segment, Country, Product, Discount Band, Units Sold, Gross Sales, Discounts, Sales, COGS, Profit and Date.
- Sales is gross sales minus discounts, and Profit is Sales minus COGS (cost of goods sold).
- Segments: Government, Small Business, Enterprise, Midmarket and Channel Partners. Countries: Canada, France, Germany, Mexico and the United States of America.

2. Data Cleaning (Python)

Issue found	How it was handled
The column name " Sales" had a leading space	Removed spaces from all column names
53 rows have the Discount Band value "None", which Pandas reads as a missing value	Kept "None" as a real category meaning no discount (keep_default_na=False)
Date was stored as text (for example 1-Dec-14)	Converted to a proper date format
Missing values and duplicates	Checked: none found (0 missing values, 0 duplicate rows)
Extra columns were needed for analysis	Added Profit Margin %, Quarter and Loss Making (Yes or No), giving 19 columns

The data was also checked for consistency: Sales equals Gross Sales minus Discounts and Profit equals Sales minus COGS in every row. The cleaned file has 700 rows and is saved as *cleaned_sales_data.csv*.

3. Key Findings

3.1 Overall performance

Total sales are 118,726,350 and total profit is 16,893,702, a margin of 14.23%. 1,125,806 units were sold across 700 orders. Of these orders, 58 lost money.

3.2 Profit by segment

Segment	Sales	Profit	Orders	Margin
Government	52,504,261	11,388,173	300	21.7%
Small Business	42,427,918	4,143,168	100	9.8%
Enterprise	19,611,694	-614,546	100	-3.1%
Midmarket	2,381,883	660,103	100	27.7%
Channel Partners	1,800,594	1,316,803	100	73.1%

- Government is the main profit driver (67% of total profit) and also has the most sales.
- Enterprise sells 19.61M but makes a loss of 614,546.
- Channel Partners has a 73% margin, the highest of all segments, but only 100 orders.

3.3 Discounts and profit margin

Discount band	Sales	Discounts given	Profit	Orders	Margin
None	7,943,654	0	1,736,455	53	21.9%
Low	34,629,779	885,676	6,188,858	160	17.9%
Medium	38,780,431	3,002,546	5,579,523	242	14.4%
High	37,372,487	5,317,026	3,388,867	245	9.1%

The pattern is clear: the bigger the discount, the lower the margin, falling from 21.9% (no discount) to 9.1% (high discount). High-discount orders gave away 5.32M in discounts.

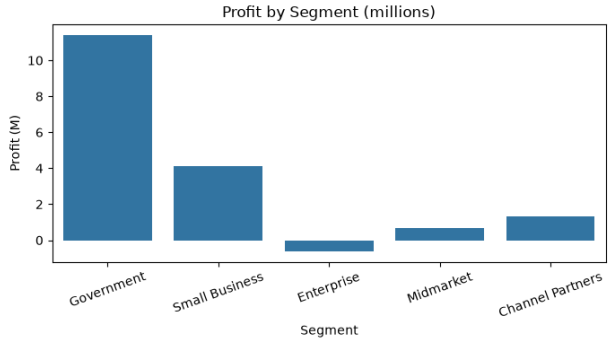


Figure 1: Profit by segment (millions)

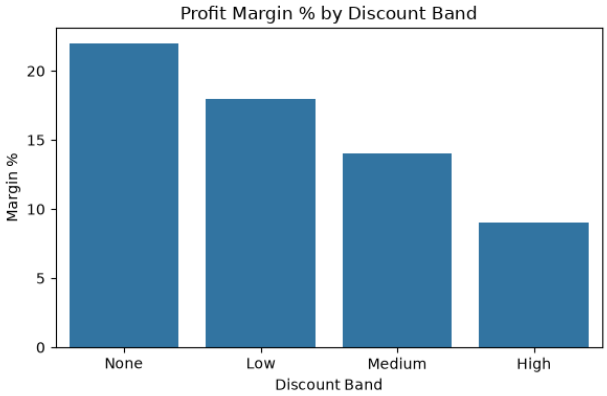


Figure 2: Profit margin % by discount band

3.4 Where the losses come from

Segment	Discount band	Orders	Loss-making orders	Profit of these orders
Enterprise	High	33	33	-622,369
Enterprise	Medium	25	25	-154,952
Enterprise	Low	36	0	101,776
Enterprise	None	6	0	61,000

Every one of the 58 loss-making orders is an Enterprise order with a High or Medium discount, and every Enterprise order in those two bands lost money. The total loss on these orders is 777,321. Enterprise orders with a Low or no discount are profitable (a combined profit of 162,776). This suggests that the discounts, rather than the Enterprise customers themselves, are a major factor in the losses.

3.5 Products and countries

Product	Sales	Profit	Margin
Paseo	33,011,144	4,797,438	14.5%
VTT	20,511,921	3,034,608	14.8%
Velo	18,250,059	2,305,992	12.6%
Amarilla	17,747,116	2,814,104	15.9%
Montana	15,390,802	2,114,755	13.7%
Carretera	13,815,308	1,826,805	13.2%

Country	Sales	Profit	Margin
USA	25,029,830	2,995,541	12.0%
Canada	24,887,655	3,529,229	14.2%
France	24,354,172	3,781,021	15.5%
Germany	23,505,341	3,680,389	15.7%
Mexico	20,949,352	2,907,523	13.9%

Paseo is the top product by sales (33.01M). Margins are fairly even across products (13% to 16%) and countries (12% to 16%). The USA has the highest sales but the lowest margin (12%).

3.6 Sales trend by quarter

Quarter	Sales	Profit	Orders	Margin
2013Q3	4,484,000	763,603	35	17.0%
2013Q4	21,931,255	3,114,861	140	14.2%
2014Q1	19,492,153	2,632,443	105	13.5%
2014Q2	22,693,880	3,232,378	140	14.2%
2014Q3	20,366,240	2,738,064	105	13.4%
2014Q4	29,758,822	4,412,352	175	14.8%

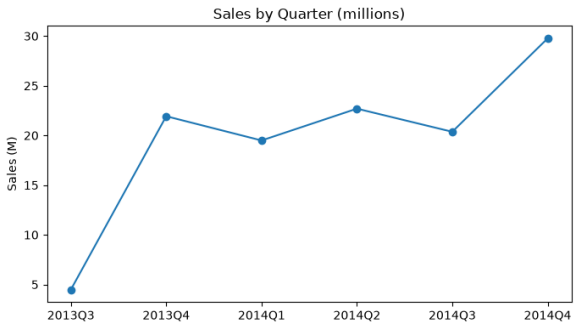


Figure 3: Sales by quarter (millions)

2014 Q4 is the strongest quarter with 29.76M in sales. The data for 2013 starts in September, so full-year 2013 (26.42M) cannot fairly be compared with full-year 2014 (92.31M). Comparing only September to December gives a fair view: 36.16M in 2014 against 26.42M in 2013, an increase of 36.9%.

4. Dashboards

Two dashboards were built from the cleaned data. Both show the same key measures: total sales, profit and margin, profit by segment, sales by quarter and profit margin by discount band, with Country and Year slicers for filtering.

4.1 Power BI dashboard

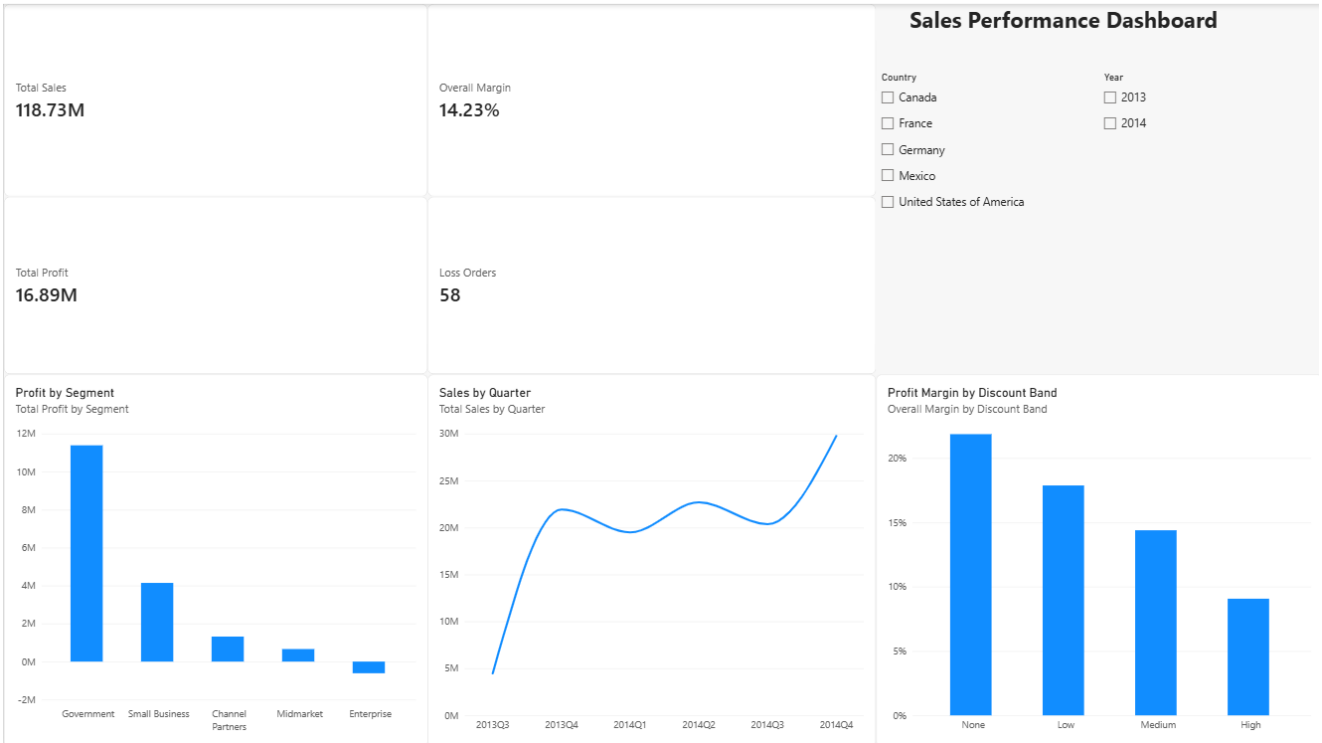


Figure 4: Power BI dashboard with KPI cards, three charts and slicers. The measures Total Sales, Total Profit, Overall Margin and Loss Orders were created with DAX.

4.2 Excel dashboard

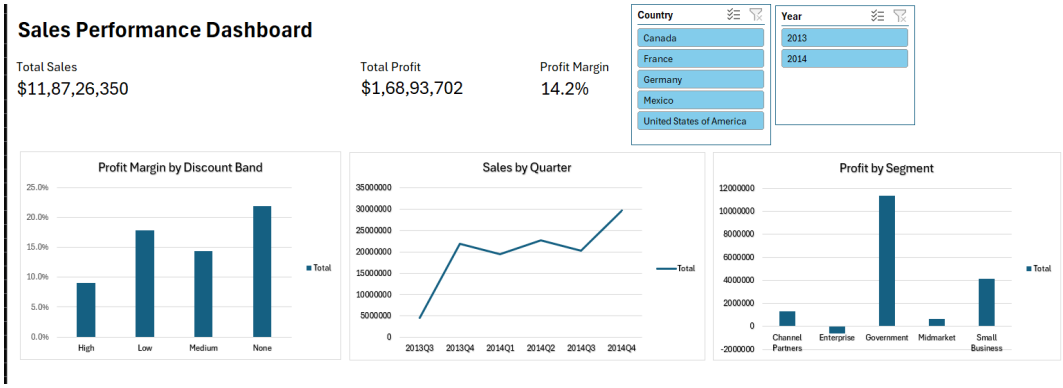


Figure 5: Excel dashboard built from pivot tables and pivot charts, with the slicers connected to every pivot table.

5. Recommendations

#	Recommendation	Evidence from the data
1	Require manager approval for Medium and High discounts on Enterprise orders.	All 58 Enterprise orders in these bands lost money (777,321 in total, 4.6% of profit).
2	Review the discount policy for all segments and set limits on High discounts.	Margin falls from 21.9% (no discount) to 9.1% (high). High discounts gave away 5.32M.
3	Test ways to grow Channel Partners and Midmarket, which are small but profitable.	Margins of 73% and 28%, but sales of only 1.80M and 2.38M.
4	Protect the Government segment and reduce dependence on it over time.	It provides 67% of total profit.
5	Prepare stock and staff for a strong fourth quarter.	2014 Q4 sales (29.76M) were the highest of any quarter, and 2013 Q4 was also strong.
6	Look into why the USA has the lowest margin despite the highest sales.	USA margin is 12% against 16% in the best countries.

Potential impact: the 58 loss-making orders cost 777,321. Even removing only these losses would raise profit by about 4.6%, before counting any extra margin from a better discount policy.

6. Limitations

- The data covers only September 2013 to December 2014, and 2013 has only four months. Trends over a longer period may differ.
- Government has 300 orders and each other segment has exactly 100, so this looks like a sample dataset. The findings apply to this dataset and should be checked against full company data.
- The dataset does not state a currency. Values are shown as they appear in the data (shown as $ in the Excel dashboard).
- The analysis shows relationships (for example, discounts and lower margins) but does not prove their cause.

7. Project Files

File	Description
capstone_analysis.ipynb	Python notebook: data cleaning, exploration, analysis and charts
sales_data.csv / cleaned_sales_data.csv	Original dataset and cleaned dataset
capstone_dashboard.xlsx	Excel pivot tables, charts and dashboard
capstone_dashboard.pbix	Power BI dashboard
Capstone_Report.pdf	This report